

Self-Supervised Pretraining for Diabetic Retinopathy Grading

(2023)

Problem

Grading diabetic retinopathy severity from retinal fundus images with limited labeled data.

Method

Contrastive self-supervised pretraining (SimCLR-style) followed by supervised fine-tuning for 5-class grading.

Dataset

EyePACS retinal fundus dataset.

Evaluation

Quadratic weighted kappa against ophthalmologist grading.

Results

Self-supervised pretraining improved quadratic weighted kappa from 0.71 (supervised-only) to 0.79.

Limitations

Cross-population generalization across different camera vendors is not evaluated in this study.